Leprosy (Hansen's Disease)

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Introduction

- Chronic infectious disease caused by Mycobacterium leprae
- Affects skin, peripheral nerves, and upper respiratory mucosa
- Long incubation period and chronic course
- Can lead to permanent disability if untreated
- Public health issue due to stigma and delayed diagnosis

EPIDEMIOLOGY

> 200,000new cases reported globally each year

Elimination target: <1 case per 10,000 population

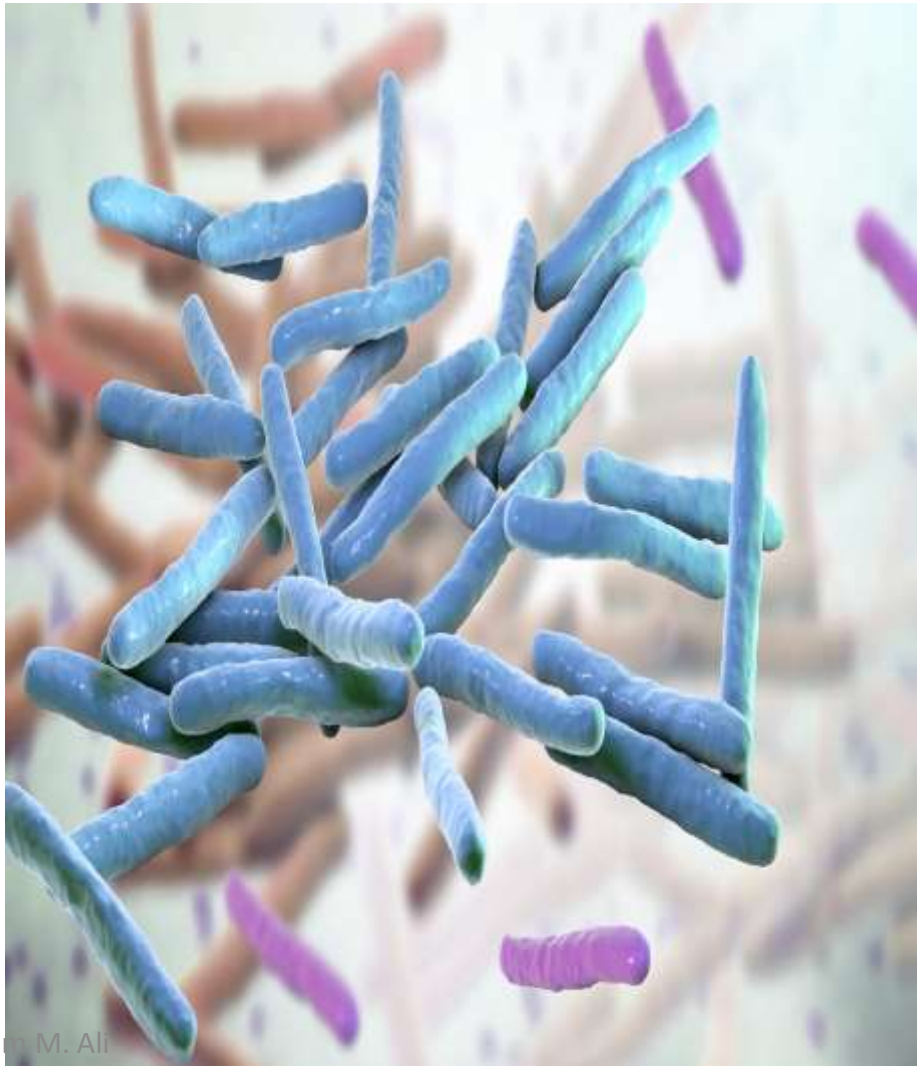
Some endemic countries still report cases

Egypt continues to report new cases annually

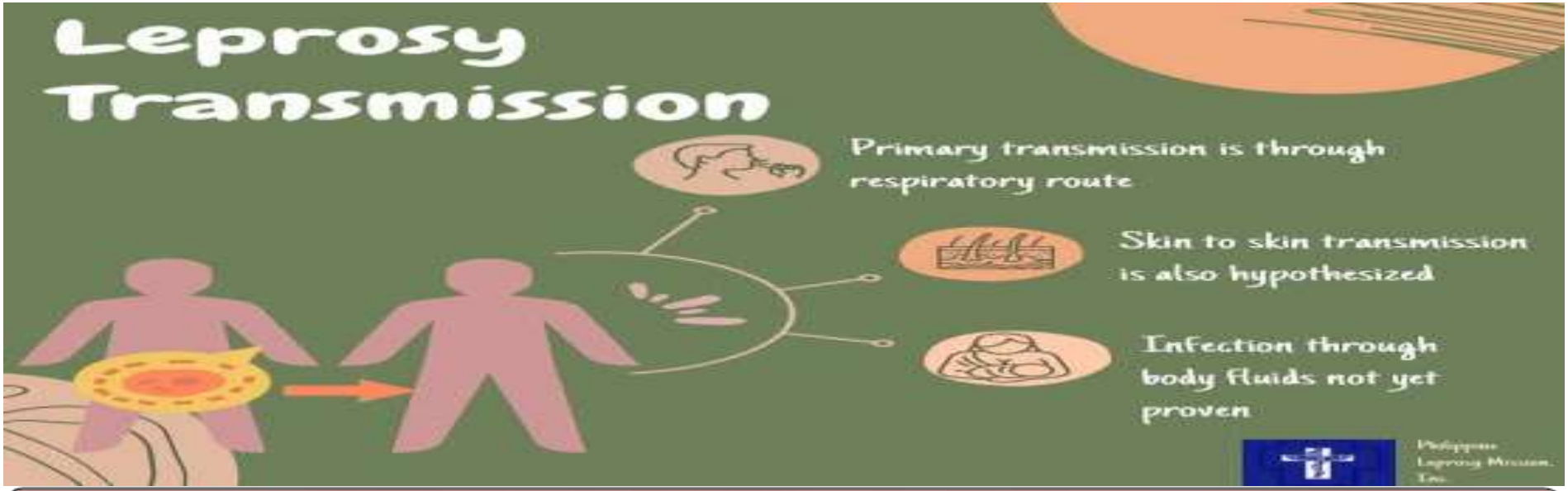
Continuous surveillance is required

ETIOLOGICAL AGENT

- Mycobacterium leprae
- Acid-fast rod-shaped bacillus
- Obligate intracellular organism
- Extremely slow multiplication



RESERVOIR & TRANSMISSION



Reservoir: **Humans**

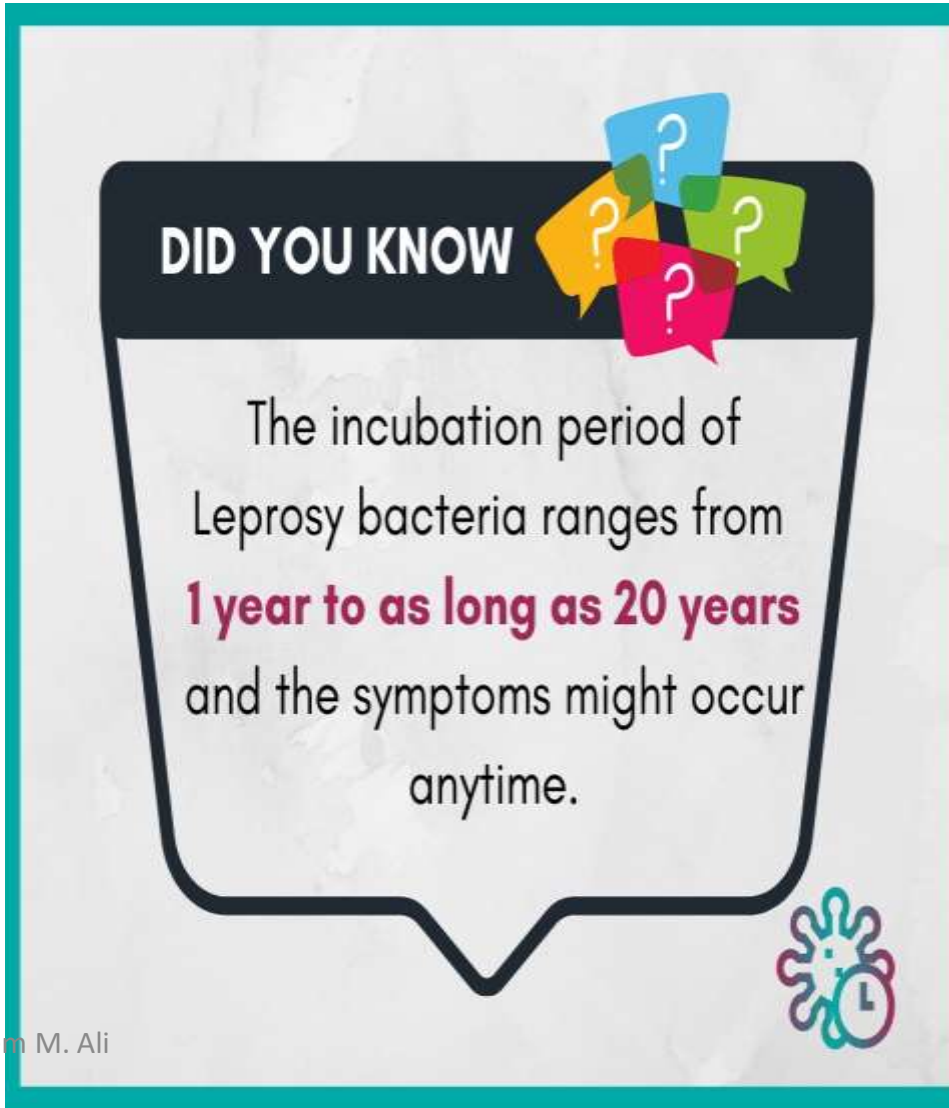
Transmission: Respiratory **droplets**

- *Close prolonged contact*
- Mainly from multibacillary cases

Dr. Tasneem M. Ali

INCUBATION PERIOD

- Range: 9months –20years
- Average 5 ≈ years
- Long incubation delays detection



CLINICAL MANIFESTATIONS

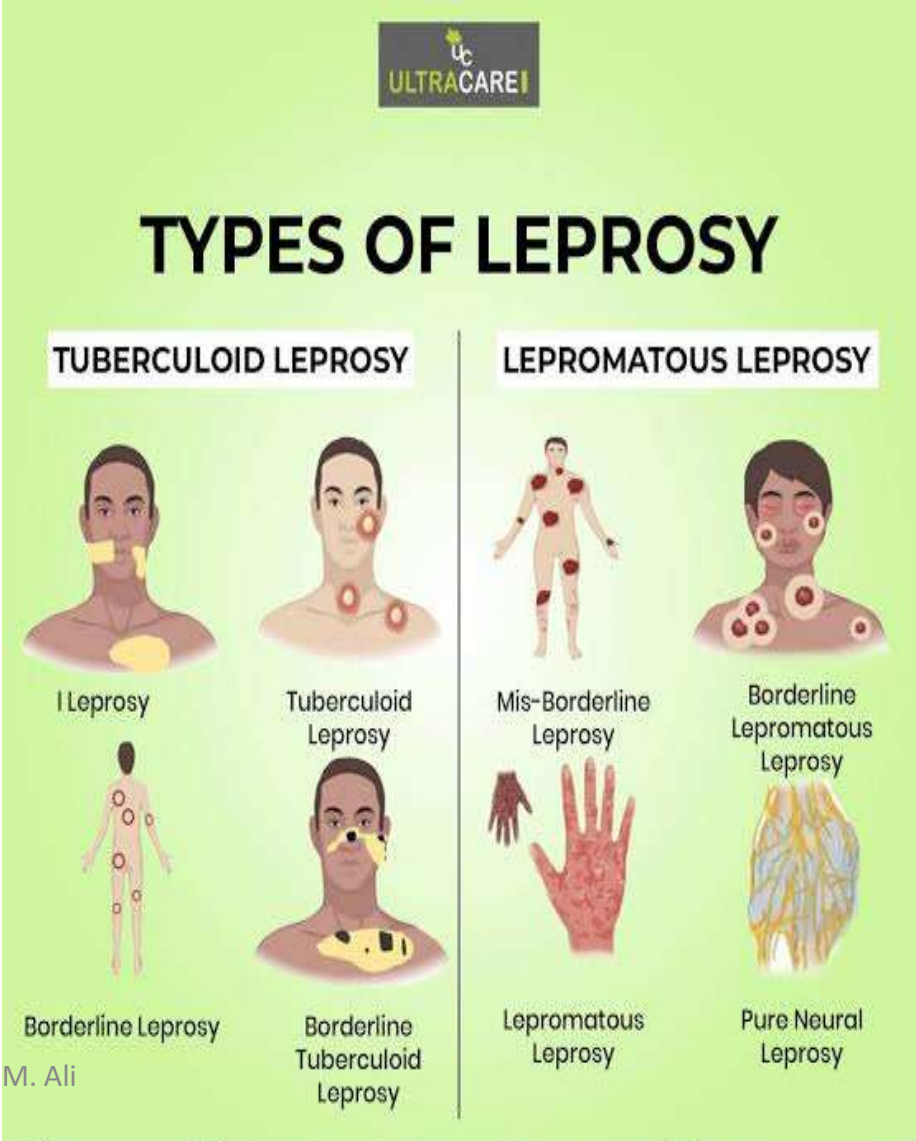
- **Skin lesions:**
- Macules
- Papules
- Nodules
- Hypopigmented lesions
- **sensory loss**

PERIPHERAL NERVE INVOLVEMENT

- Thickened peripheral nerves
- Loss of sensation
- Muscle weakness
- Possible deformities

IMMUNOLOGICAL SPECTRUM

- Tuberculoid – strong immunity
- Lepromatous – weak immunity
- Borderline – intermediate form



Feature	Lepromatous Leprosy	Tuberculoid Leprosy
Skin lesions	Numerous, extensive	Single or few
Distribution	Bilaterally symmetrical	Asymmetrical
Lesion margins	Diffuse infiltrations	Sharply demarcated
Peripheral nerve involvement	Less severe	Severe
Nasal mucosa involvement	Present	Absent
Ocular involvement	Present	Absent

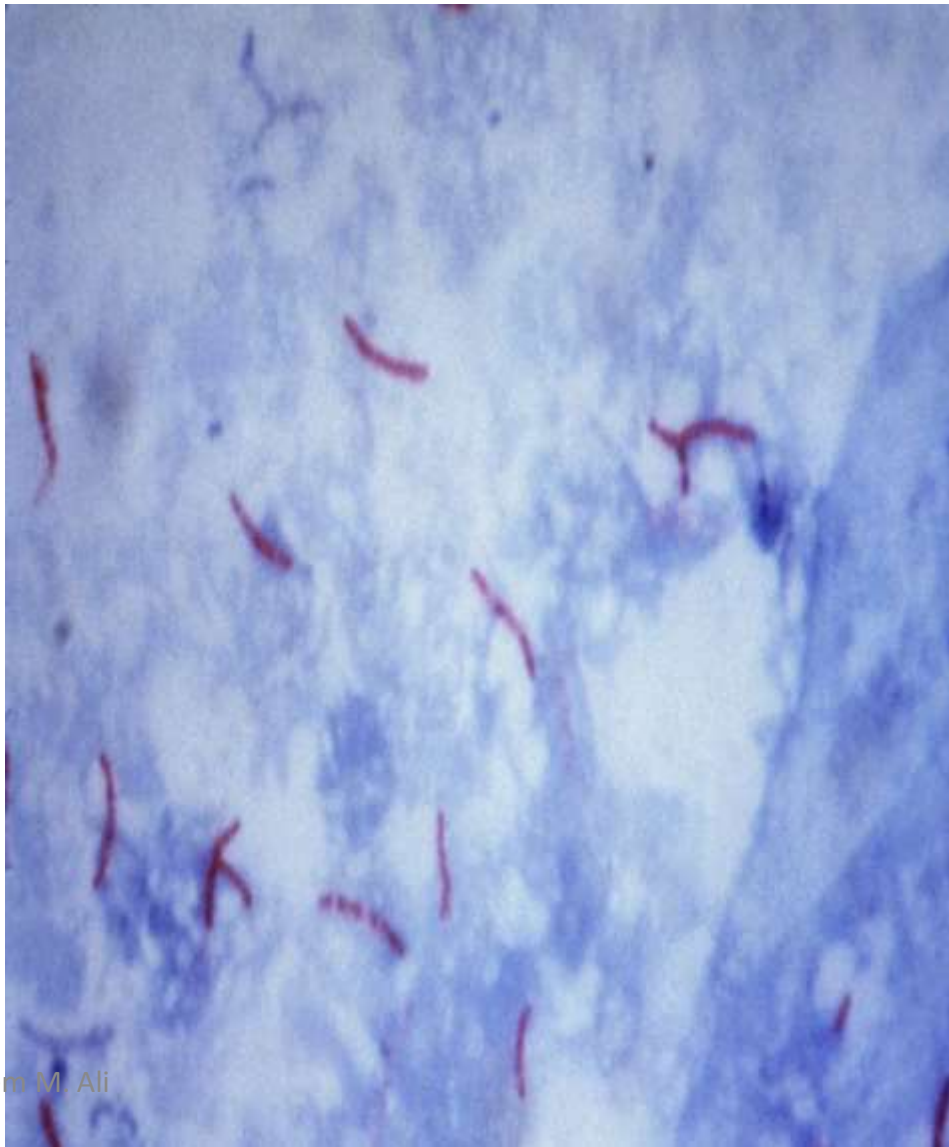
LEPROSY REACTIONS

- Reversal reaction in (BL)
- Erythema nodosum leprosum in (LL)

These reactions contribute significantly to nerve damage and disability.

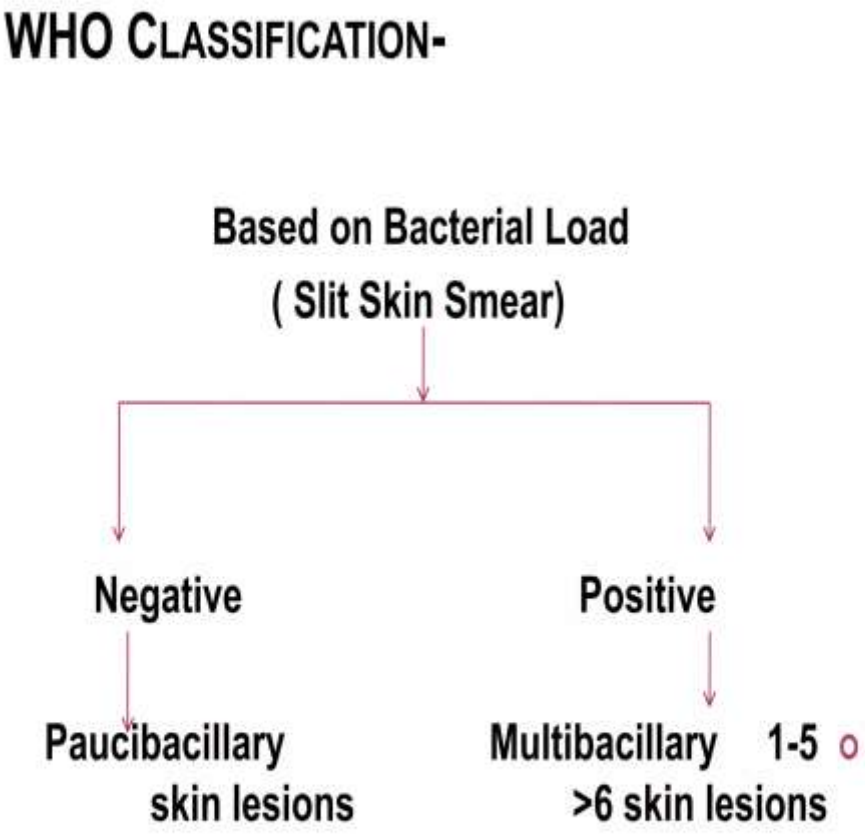
Diagnosis

- • Mainly clinical
- • Skin lesion with sensory loss
- • Skin smear may detect acid-fast bacilli



Classification

- Paucibacillary (PB): smear negative
- Multibacillary (MB): smear positive
- Clinical criteria often used



Infectivity drops after treatment

3months after dapsone/clofazimine

3 days after rifampicin

Prevention of Leprosy

- Vaccination of BCG
- Educate people about its causes

PRIMARY PREVENTION

Health education emphasizing curability, availability of effective multi-drug therapy (MDT), and reduction of stigma

BCG **vaccination** offers partial protection, particularly against tuberculoid leprosy

Early case detection and prompt MDT, especially for infectious MB cases.

SECONDARY PREVENTION

1. **Notification:** Leprosy is a notifiable disease in many countries

2. **Isolation: Not required for tuberculoid leprosy, Temporary contact isolation for lepromatous cases until MDT is established**

3. **Disinfection:** Concurrent disinfection of nasal secretions and terminal cleaning are recommended.

4. **Contact Surveillance:** Household and close contacts should be examined initially and then **annually for at least 5 years** after last exposure.

5. **Specific Treatment (Multi-Drug Therapy – MDT):** MDT is the cornerstone of leprosy control. Monotherapy is unethical due to inevitable drug resistance.

TREATMENT (MDT)

- Drugs:
- • Rifampicin Most potent bactericidal drug; administered once monthly
- • Clofazimine Causes reversible skin discoloration
- • Dapsone Generally safe; rare allergic reactions may occur

- Combination therapy prevents resistance

WHO Regimens

- • MB: Rifampicin + Dapsone + Clofazimine (24 months)
- • PB: Rifampicin + Dapsone (6 months)
- • Monthly supervised doses

MDT Regimens

Each blister pack contains treatment for 4 weeks.

PB adult blister pack

PB adult treatment:

Once a month: Day 1
– 2 capsules of rifampicin (300 mg X 2)
– 1 tablet of dapsone (100 mg)

Once a day: Days 2–28
– 1 tablet of dapsone (100 mg)

Full course: 6 blister packs

MB adult blister pack

MB adult treatment:

Once a month: Day 1
– 2 capsules of rifampicin (300 mg X 2)
– 3 capsules of clofazimine (100mg X 3)
– 1 tablet of dapsone (100 mg)

Once a day: Days 2–28
– 1 capsule of clofazimine (50 mg)
– 1 tablet of dapsone (100 mg)

Full course: 12 blister packs

Conclusion

- Leprosy is curable
- Early detection prevents disability
- MDT is essential for control
- Continued surveillance and education are needed

- **1. Household Contact Examination**
- **Frequency:** Annually for at least 5 years
- **Purpose:** Early detection of new cases among close contacts
- **2. Vaccination**
- **BCG vaccine** provides partial protection against leprosy
- **3. Public Health Control Strategy**
- **Most effective approach:**
  - Early detection of cases
  - Prompt initiation of **Multidrug Therapy (MDT)**
- **Goal:** Reduce community transmission
- **Tips to Remember:**
- Contacts need **long-term follow-up**, not just when symptoms appear
- MDT is preferred over isolation or mass prophylaxis
- BCG helps, but does **not replace treatment or contact surveillance**

SHOW What You KNOW

A Quick Way to Check for Understanding!

THE CLASSROOM NOOK

A rural district reports a leprosy prevalence of 0.8 per 10,000 population but continues to detect high numbers of new multibacillary (MB) cases annually. Which interpretation best explains the situation?

A. The district has successfully eliminated leprosy as a public health problem and transmission has stopped.

B. Low prevalence means the district should stop active surveillance and focus only on treatment.

C. The low prevalence reflects effective MDT outcomes, but the high MB detection indicates ongoing transmission.

D. The high MB detection indicates that leprosy is becoming more contagious over time.

A family member has lived with an untreated MB leprosy patient for several years. Which factor most strongly determines whether the family member will develop clinical disease?

- A. Duration of exposure only
- B. The virulence of Mycobacterium leprae
- C. Level of the household contact's cell-mediated immunity
- D. Frequency of disinfection in the home

A health team must choose the most effective strategy to rapidly reduce community infectiousness in an endemic area. Which action will produce the fastest impact?

- A. Expanding BCG vaccination to all school-aged children
- B. Treating all confirmed MB cases with rifampicin-containing MDT immediately
- C. Annual screening of contacts for 5 years
- D. Hospitalizing all lepromatous patients for isolation

During a civil conflict, many patients default from MDT for months. What is the most likely epidemiological consequence?

- A. Sharp decrease in new cases due to reduced mobility
- B. Increase in drug resistance and rising community transmission
- C. Fewer MB cases because untreated disease becomes self-limiting
- D. Reduced nerve damage due to interruption of therapy

A newly diagnosed MB patient starts MDT today. The family asks when the patient stops being infectious. Based on the document, which response is most accurate?

- A. The patient remains infectious for at least 6 months after starting MDT.
- B. Infectiousness stops only after completing the full 24-month MB regimen.
- C. Infectiousness is markedly reduced within 3 days of receiving rifampicin.
- D. The patient is noninfectious immediately after the first dose of dapsone alone.

感謝你
Danke
Gracias
Grazie
Merci
Thanks!
THANK YOU!